

Discrete Structures

Practice Problem Set 5

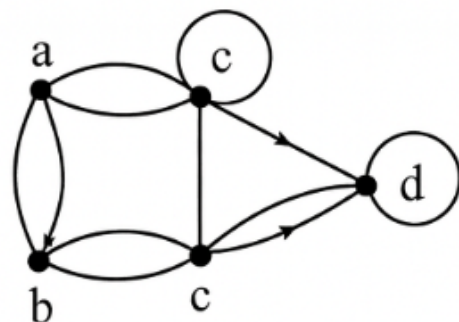
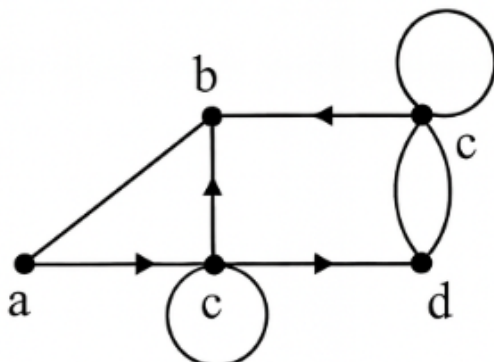
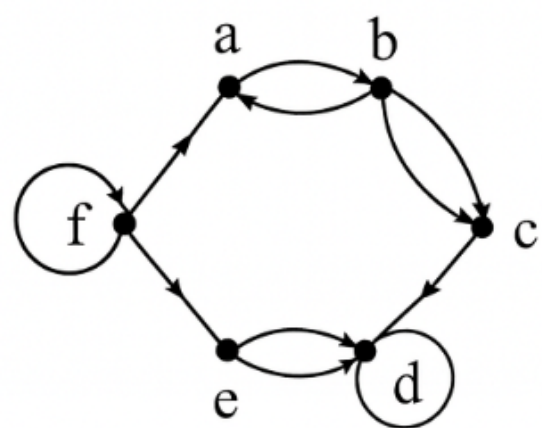
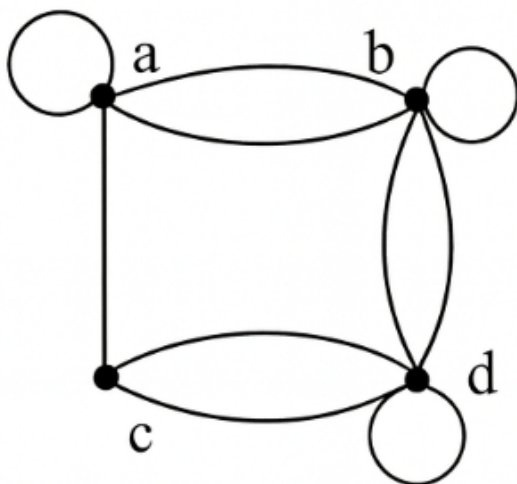
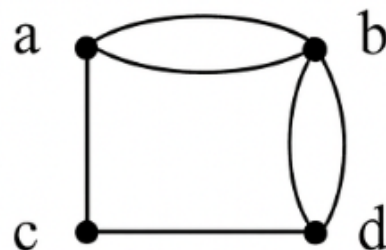
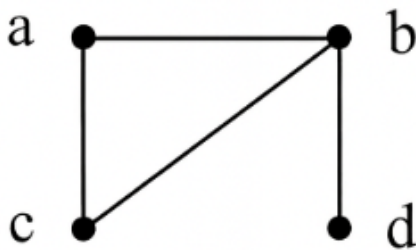
Release Date: November 18th, 2025

Instructions:

- These problems are meant only for your practice.
- Submission is not required.

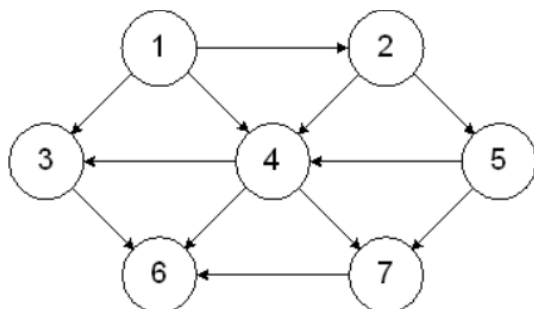
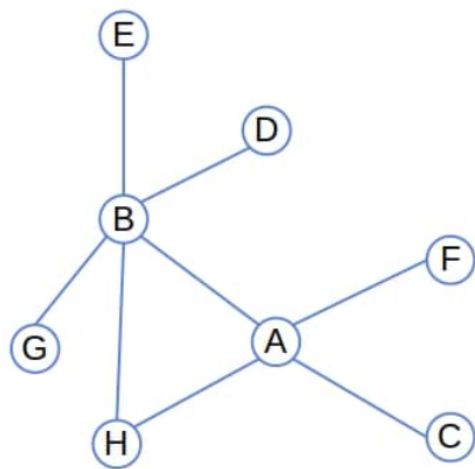
Problem 1:

Classify the graph type e.g., Simple, Multi, Pseudo * Directed, Undirected for the given graphs.



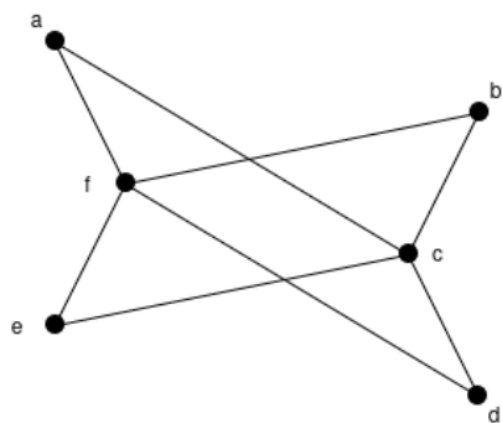
Problem 2:

Compute degree and neighbors the following graphs

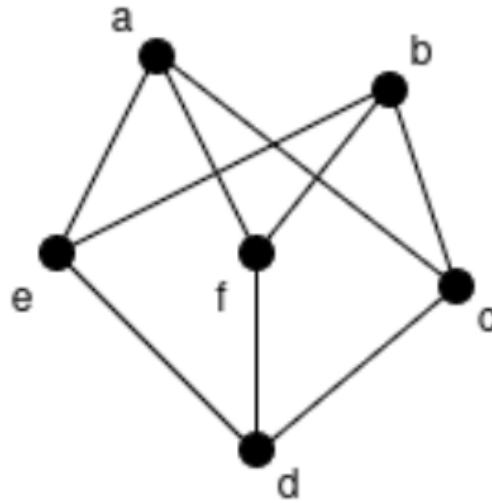


Problem 3:

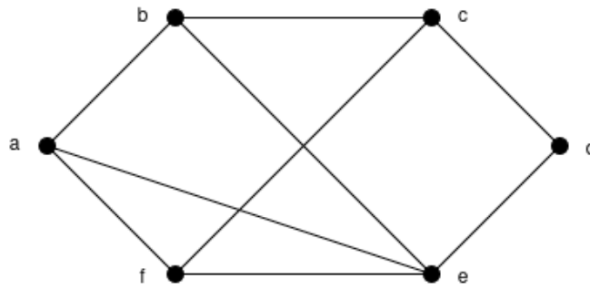
Given the following graphs, determine whether each graph is bipartite or not.



(a)



(b)



(c)

Problem 4:

Consider the graph $G = (V, E)$ with

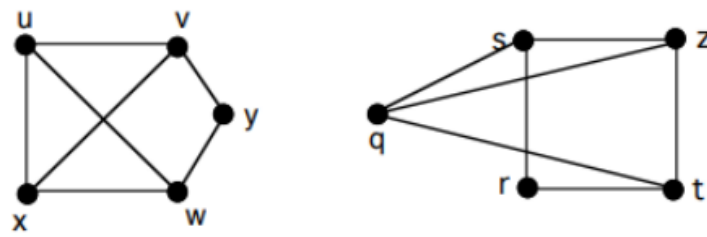
$$V = \{1, 2, 3, 4, 5, 6\}, \quad E = \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 5), (4, 5), (5, 6)\}.$$

Let the vertex set $S = \{1, 2, 3, 5\}$. Then answer the following.

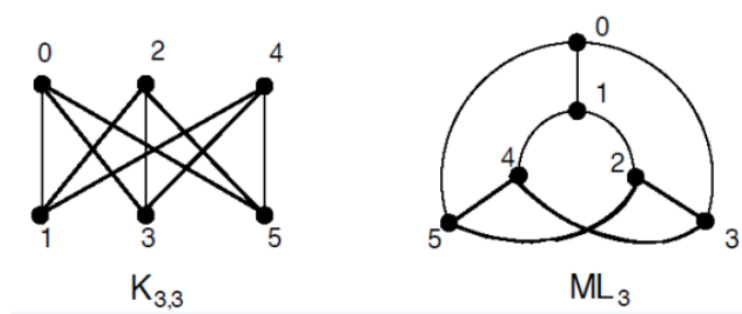
- (a) List all edges of the induced subgraph $G[S]$. Draw (or sketch) $G[S]$ below.
- (b) You are shown the following candidate subgraph H_1 on vertex set S :
 $E(H_1) = \{(1, 2), (3, 5)\}$.
 Is H_1 the induced subgraph $G[S]$? If not, list which edges (if any) are missing from H_1 to make it equal to the induced subgraph.
- (c) Another graph H_2 has edges $E(H_2) = \{(1, 3), (2, 3), (3, 5)\}$. Determine whether there exists a vertex set $T \subseteq V$ such that H_2 is an induced subgraph $G[T]$. If yes, find such a T . If no, explain which defining property of induced subgraphs is violated.

Problem 5

Identify if the below pairs of graphs are isomorphic or not. If yes, find an isomorphism.



(a)



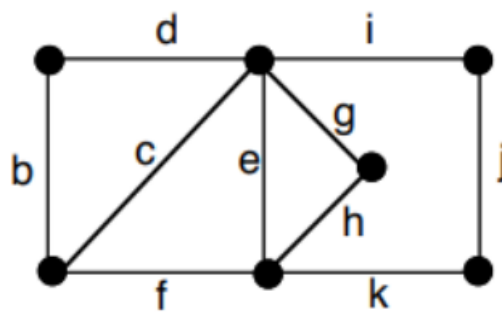
(b)



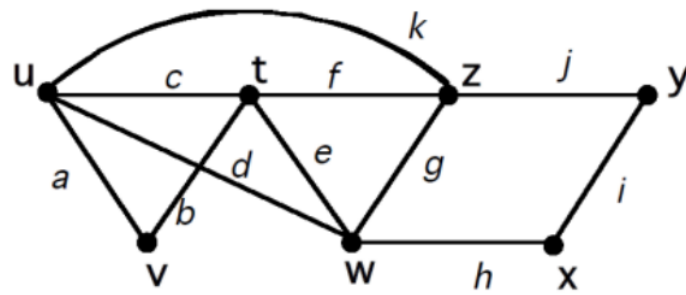
(c)

Problem 6:

(a) Does the graph have an Euler path? If yes, find one.



(b) Does this graph have an Euler circuit? If yes, find one.



(c) Consider the directed graph shown below

- Does the above graph have a directed Euler circuit? Justify your answer using in-degree and out-degree conditions.
- Does the above graph have a directed Euler path (but not a circuit)? State the necessary conditions and verify them for this graph.
- If an Euler circuit or path exists, construct one explicitly by listing the sequence of vertices in order.

